



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of India)
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Symbiosis Institute of Technology, Pune

Faculty of Engineering

CSE- Academic Year 2026-27

Java Programming – Lab Batch 2025-29

Lab Assignment No:- 20	
Name of Student	RAM SHARMA
PRN No.	25070122150
Batch	CSE-B2
Class	PROGRAMMING WITH JAVA
Academic Year & Semester	2nd-year and 3rd-semester
Date of Submission	04/10/2026
Title of Assignment:	Develop a Java application to establish connectivity with a database and perform INSERT,UPDATE, and DELETE operations.
Problem Statement	Develop a Java application to establish connectivity with a database and perform INSERT,UPDATE, and DELETE operations.
Learning objective	To perform database modification operations using JDBC.
Theory	JDBC allows Java applications to perform database manipulation using SQL DataManipulation Language (DML) statements such as INSERT, UPDATE, and DELETE. TheINSERT statement adds new records into database tables, UPDATE modifies existing records, and DELETE removes unwanted records. These operations are executed using the Statement or PreparedStatement interface. PreparedStatement is generally preferred

	because it supports parameterized queries, improves performance, and prevents SQL injection attacks. Proper transaction management ensures that database modifications are performed safely and consistently. These operations form the basis of almost every database-driven application such as banking systems, hospital management systems, and student information systems.
Programme	1. Create a Java application to manage employee records using CRUD operations =>import java.sql.*; <pre> public class E1 { public static void main(String[] args) { String url = "jdbc:mysql://localhost:3306/jdbc_demo"; String user = "root"; String password = "YOUR_MYSQL_PASSWORD"; try { // Load JDBC driver Class.forName("com.mysql.cj.jdbc.Driver"); // Connect to database Connection con = DriverManager.getConnection(url, user, password); System.out.println("Database connected successfully.\n"); // CREATE - Insert employee String insertQuery = "INSERT INTO employee VALUES (?, ?, ?, ?, ?)" PreparedStatement ps1 = con.prepareStatement(insertQuery); ps1.setInt(1, 104); ps1.setString(2, "Rohit"); ps1.setInt(3, 24); ps1.setString(4, "Delhi"); ps1.setDouble(5, 40000); ps1.executeUpdate(); </pre>

```
System.out.println("Employee inserted successfully.");

// READ - Display employees
String selectQuery = "SELECT * FROM employee";

Statement st = con.createStatement();
ResultSet rs = st.executeQuery(selectQuery);

System.out.println("\nEmployee Records:");

while (rs.next()) {

    System.out.println(
        rs.getInt("id") + " " +
        rs.getString("name") + " " +
        rs.getInt("age") + " " +
        rs.getString("address") + " " +
        rs.getDouble("salary")
    );
}

// UPDATE - Update salary
String updateQuery = "UPDATE employee SET salary = ? WHERE
id = ?";

PreparedStatement ps2 = con.prepareStatement(updateQuery);

ps2.setDouble(1, 45000);
ps2.setInt(2, 104);

ps2.executeUpdate();

System.out.println("\nEmployee salary updated successfully.");

// DELETE - Delete employee
String deleteQuery = "DELETE FROM employee WHERE id = ?";

PreparedStatement ps3 = con.prepareStatement(deleteQuery);

ps3.setInt(1, 103);

ps3.executeUpdate();
```

```

        System.out.println("Employee deleted successfully.");

        // Close connection
        con.close();

    } catch (Exception e) {

        System.out.println("Error: " + e.getMessage());
    }
}
}

```

2. Create a Java application to manage **student records** using **CRUD operations**. Perform Create, Read, Update, and Delete operations on student details such as roll number, name, course, and marks using a database.

=>import java.sql.*;

```

public class E2 {

    public static void main(String[] args) {

        String url = "jdbc:mysql://localhost:3306/jdbc_demo";
        String user = "root";
        String password = "YOUR_MYSQL_PASSWORD";

        try {

            // Load JDBC driver
            Class.forName("com.mysql.cj.jdbc.Driver");

            // Connect to database
            Connection con = DriverManager.getConnection(url, user,
password);

            System.out.println("Database connected successfully.\n");

            // CREATE - Insert student
            String insertQuery =
                "INSERT INTO student VALUES (?, ?, ?, ?)";

            PreparedStatement ps1 =
                con.prepareStatement(insertQuery);

```

```
ps1.setInt(1, 4);
ps1.setString(2, "Rohit");
ps1.setString(3, "CSE");
ps1.setDouble(4, 82);

ps1.executeUpdate();

System.out.println("Student inserted successfully.");

// READ - Display students
String selectQuery = "SELECT * FROM student";

Statement st = con.createStatement();

ResultSet rs = st.executeQuery(selectQuery);

System.out.println("\nStudent Records:");

while (rs.next()) {

    System.out.println(
        rs.getInt("rollno") + " " +
        rs.getString("name") + " " +
        rs.getString("course") + " " +
        rs.getDouble("marks")
    );
}

// UPDATE - Update marks
String updateQuery =
    "UPDATE student SET marks = ? WHERE rollno = ?";

PreparedStatement ps2 =
    con.prepareStatement(updateQuery);

ps2.setDouble(1, 90);
ps2.setInt(2, 4);

ps2.executeUpdate();

System.out.println("\nStudent marks updated successfully.");
```

```

// DELETE - Delete student
String deleteQuery =
    "DELETE FROM student WHERE rollno = ?";

PreparedStatement ps3 =
    con.prepareStatement(deleteQuery);

ps3.setInt(1, 3);

ps3.executeUpdate();

System.out.println("Student deleted successfully.");

// Close connection
con.close();

} catch (Exception e) {

    System.out.println("Error: " + e.getMessage());
}
}
}

SQL:
USE jdbc_demo;

-- =====
-- E1: EMPLOYEE TABLE
-- =====

DROP TABLE IF EXISTS employee;

CREATE TABLE employee (
    id INT PRIMARY KEY,
    name VARCHAR(50),
    age INT,
    address VARCHAR(100),
    salary DOUBLE
);

INSERT INTO employee VALUES
(101, 'Rahul', 22, 'Pune', 30000),
(102, 'Amit', 23, 'Mumbai', 35000),

```

```
(103, 'Priya', 22, 'Nashik', 32000);
```

```
SELECT * FROM employee;
```

```
-- =====  
-- E2: STUDENT TABLE  
-- =====
```

```
DROP TABLE IF EXISTS student;
```

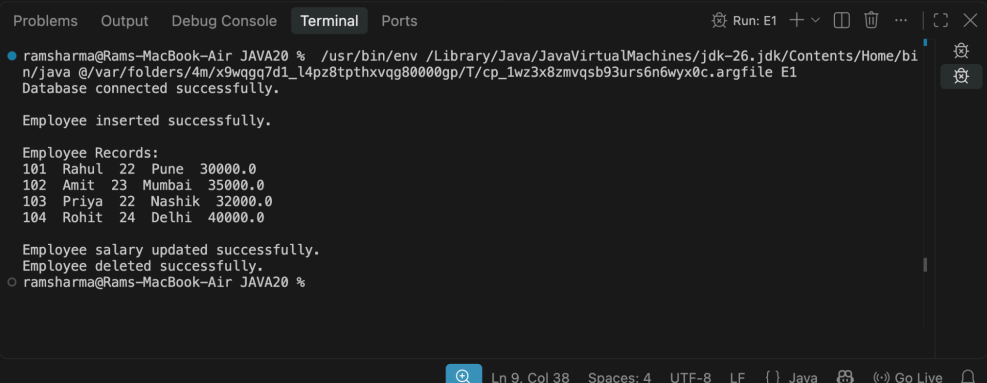
```
CREATE TABLE student (  
    rollno INT PRIMARY KEY,  
    name VARCHAR(50),  
    course VARCHAR(50),  
    marks DOUBLE  
);
```

```
INSERT INTO student VALUES  
(1, 'Rahul', 'CSE', 85),  
(2, 'Amit', 'CSE', 78),  
(3, 'Priya', 'ECE', 88);
```

```
SELECT * FROM student;
```

Output Screenshots (if applicable)

E1 =>



```
Problems Output Debug Console Terminal Ports
Run: E1 + - [] ... | [X] X
ramsharma@Rams-MacBook-Air JAVA20 % /usr/bin/env /Library/Java/JavaVirtualMachines/jdk-26.jdk/Contents/Home/bin/java @/var/folders/4m/x9wqgq7d1_l4pz8tphxvqg80000gp/T/cp_1wz3x8zmqsb93urs6n6wyx0c.argfile E1
Database connected successfully.

Employee inserted successfully.

Employee Records:
101 Rahul 22 Pune 30000.0
102 Amit 23 Mumbai 35000.0
103 Priya 22 Nashik 32000.0
104 Rohit 24 Delhi 40000.0

Employee salary updated successfully.
Employee deleted successfully.
ramsharma@Rams-MacBook-Air JAVA20 %
```

Ln 9, Col 38 Spaces: 4 UTF-8 LF {} Java Go Live

```
Problems Output Debug Console Terminal Ports
● ramsharma@Rams-MacBook-Air JAVA20 % /usr/bin/env /Library/Java/JavaVirtualMachines/jdk-26.jdk/Contents/Home/bin/java @var/folders/4m/x9wqqg7d1_l4pz8tphxvqg80000gp/T/cp_1wz3x8zmvsq93urs6n6wy0c.argfile E2
Database connected successfully.

Student inserted successfully.

Student Records:
1 Rahul CSE 85.0
2 Amit CSE 78.0
3 Priya ECE 88.0
4 Rohit CSE 82.0

Student marks updated successfully.
Student deleted successfully.
○ ramsharma@Rams-MacBook-Air JAVA20 %
```

The screenshot displays the GitHub web interface for the repository 'RamSharma-19 / JAVA-LAB-ASSIGNMENTS'. The 'main' branch is selected. The file explorer on the left shows a directory structure with folders JAVA10 through JAVA20 and a 'lib' folder. The main content area shows the 'lib' folder selected, displaying a table of files: 20.sql, E1.java, and E2.java, all with the commit message 'Add files via upload' and the date 'now'.

In this assignment, JDBC was used to perform CRUD operations on employee and student records. The programs demonstrated INSERT, SELECT, UPDATE, and DELETE operations using SQL queries. Connection, Statement, PreparedStatement, and ResultSet were used to interact with the MySQL database. The employee and student records were successfully added, displayed, modified, and deleted. This experiment helped in understanding how Java applications can manage and update data stored in a relational database.

